

A Study of the Accuracy of Artificial Intelligence in the English Translation of Some Nigerian Indigenous Language Expressions

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Abstract

Machine Translation Tools (MTT) such as Google Translate, DeepL Translator, OpenAI GPT have become widely used for various translation tasks. While these tools offer speed, accessibility, and cost-effectiveness, there are concerns regarding accuracy, contextual appropriateness, and stylistic nuances, particularly where African indigenous languages that manifest unique linguistic structures are concerned. Therefore, this paper examined the extent to which MTT could aid or hinder the translation quality of indigenous languages by investigating their accuracy with regard to lexical choice, syntactic structure, proverbs, idiomatic expressions, and cultural nuances. Data comprised fifteen (15) purposively selected sentences from Igbo, Hausa and Yoruba languages. These sentences were gathered from the fields of education, healthcare, legal, business and international diplomacy. The sentences were then subjected to MTT and Human Translation (HT) in order to determine their appropriateness in light of the Nigerian cultural context in which they are used. Specimen sentences were drawn from the English language; these specimen sentences were translated into the indigenous languages called the source sentences. These source sentences were then subjected to MTT and HT. The translated sentences were evaluated with the specimen sentences. Findings indicate that while MTT have significantly improved in precision, they still struggle with domain-specific terminology and pragmatic subtleties, necessitating intervention for high-quality translations. The study underscores the need for a hybrid approach, combining MTT efficiency with HT expertise, to achieve optimal translation quality in professional and academic contexts.

Keywords: Machine translation tools (MTT), human translation (HT), indigenous languages, linguistic accuracy, artificial intelligence.

Introduction

In the modern digital age, the advent of machine translation tools (MTT) has revolutionized the way languages are translated, bridging communication gaps across linguistic boundaries and offering rapid and accessible solutions to multilingual communication. However, the quality of translation produced by MTT, particularly in English, remains a critical area of inquiry. The rapid advancement of artificial intelligence tools such as Google Translate, DeepL Translator, Online Dictionary, Microsoft Translator, and Open AI GPT to mention but a few, have become ubiquitous in various fields. Artificial Intelligence (AI) refers to the simulation of human intelligence in machines that are programmed to think and learn like humans (Sheikh, 2023). AI could be said to be a discipline focused on designing and building different machines capable of performing tasks that require intelligence, similar to human thinking (McCarthy, 2007; Nilsson, 1998). Others (e.g. Mackworth & Goebel 1998; Sheikh, 2023) posit that AI examines the kind of computational processes that allow machines to perceive, reason,

and take meaningful actions. Despite their efficiency and accessibility, the quality of translations provided by these tools have been a subject of debate, particularly in ensuring linguistic accuracy, contextual understanding, cultural appropriateness and sensitivity. MTT (AI Applications) is a subfield of Applied Linguistics. Linguistics itself is the scientific study of language (Allan, 2015). This study seeks to examine the impact of MTT on the quality of English translations within the domain of applied linguistics. The study evaluates the performance of MTT against key quality benchmarks, including lexical choice, syntax, idiomatic expressions through comparative analyses of MTT and HT. Furthermore, it addresses the broader sociolinguistic and ethical implications of relying on MTT, particularly in critical domains such as education, healthcare, law, business and international diplomacy.

Translation has long been a vital tool for fostering communication, preserving knowledge, and bridging cultural divides. Historically, translation relied exclusively on human effort (HT), involving skilled linguists who meticulously rendered meaning from one language to another (Bowker, 2023). However, the rise of Digital Technology and AI has transformed this landscape, introducing MTT that automate the translation process. These tools, powered by sophisticated algorithms, have become widely accessible and are increasingly used for a variety of purposes, from casual conversation to professional documentation. MTT has evolved through several stages, beginning with rule-based systems (RB) in the mid-20th century to the more recent neural machine translation (NMT) models (Bowker, 2002). NMT, which underpins modern tools like Google Translate, DeepL Translator, OpenAI GPT and Microsoft translator, leverages deep learning techniques to produce more fluent and contextually appropriate translations. Despite these advancements, questions about the quality and reliability of machine-generated translations remain, particularly when the target language is English. The complexity of translating into English is multifaceted, involving not only grammatical accuracy but also the preservation of proverbs, idiomatic expressions, cultural nuances, and contextual meaning. While MTT can produce rapid translations, their ability to fully replicate the intricacies of human language remains a contentious issue. English, as a global language, requires translations that are not only grammatically accurate but also contextually appropriate and culturally sensitive. Many studies highlight the strengths of MTT, such as speed and cost-efficiency (Bowker, 2019), but less emphasis has been placed on their limitations in handling the subtleties of meaning, idiomatic expressions and cultural nuances.

Therefore, the objectives of this study are to: (i) examine the use of MTT in the translation of the selected sentences, (ii) analyse the sentences to determine the accuracy, fluency and grammatical correctness of English translations produced by MTT, and (iii) discuss the implications of machine translation on the future role of professional human translators. The study is considered significant because it has the potential to contribute to a deeper understanding of the impact of MTT on the quality of English translation of African languages in general, and Nigerian indigenous languages in particular. In addition, it is expected to provide insight into the practical effectiveness of MTT, identifying areas that could enhance the technology and practice of translation in a rapidly evolving digital world.

Literature Review

Conceptual Review

Machine Translation Tools (MTT) refer to computer-based systems designed to automatically translate text or speech from one language to another. These tools rely on computational algorithms, linguistic databases, and increasingly AI to process language structures and generate equivalent meanings in the target language. Early machine translation systems were rule-based, relying on preprogrammed grammatical rules and bilingual dictionaries. However, modern systems are largely based on statistical and neural models that learn patterns from large corpora of multilingual data. According to Manning and Schütze (1999), machine translation systems analyze linguistic patterns within large datasets to generate probable equivalents in another language. This process involves tokenization, syntactic parsing, and semantic modeling. Similarly, Koehn (2010) explains that modern translation systems have moved from rule-based models to statistical and neural approaches that improve translation fluency and contextual interpretation. The integration of neural networks in translation technology has led to the development of Neural Machine Translation (NMT) systems that provide more contextually appropriate translations. Bengio, Hinton, and LeCun (2015) have contributed significantly to the advancement of deep learning methods that power these translation systems. Despite these improvements, MTT still faces challenges when translating low-resource languages, especially indigenous languages that lack large digital corpora. Linguistic nuances, cultural expressions, and idiomatic constructions are often difficult for automated systems to capture accurately.

Human Translation (HT) refers to the process whereby professional translators or language experts interpret and convert text from a source language into a target language while preserving meaning, context, and cultural nuance. Unlike machine translation, human translators apply linguistic knowledge, contextual awareness, and cultural understanding to ensure that translations are meaningful and appropriate. Nida (1964) introduced the concept of dynamic equivalence, emphasizing that translation should reproduce the same effect on the target audience as the original text had on its audience. This perspective highlights the importance of context and cultural interpretation in translation. Similarly, Newmark (1988) distinguishes between semantic translation, which focuses on literal accuracy, and communicative translation, which prioritizes the intended message and audience comprehension. Human translators are capable of balancing these approaches depending on the context. Another scholar, Baker (2018), argues that translation is not merely linguistic substitution but a complex process involving discourse analysis, pragmatics, and sociocultural interpretation. This complexity explains why human translation often produces more culturally appropriate results than machine-generated translations. Therefore, human translation provides a benchmark for evaluating the effectiveness of machine translation tools.

MTT refers to the use of software to translate text or speech from one language to another. Over time, MTT has evolved from Rule-based system (RBMT) to sophisticated non-rule-based methods (Non-RBMT), including statistical and neural machine translation. The table below shows the differences between RBMT and Non-RBMT.

Table 1: Key Differences Between RBMT and Non-RBMT

Features	RBMT	Non-RBMT
Approach	Rule-based relies on linguistic rules; predefined linguistic rules, grammar, and dictionaries	Uses data-driven methods like statistics or neural networks, learns from bilingual corpora or patterns.
Flexibility	Limited to the quality of the rules and lexicons provided, requires manual rule updates.	Highly flexible, adaptable and improves with sufficient and availability of larger datasets.
Linguistic Knowledge	Requires extensive linguistic expertise and manual rule creation	Does not require explicit linguistic rules; learns patterns from data.
Contextual Understanding	Poor; struggles with idiomatic text.	Better; excels in context and idiomatic phrases.
Accuracy for Low-Resource Languages	High if rules exist for language pair	Low without sufficient data.
Scalability	Difficult due to manual effort.	Scalable with computational resources.
Output Quality	More literal and structured; often lacks natural fluency	More fluent and context-aware; however, it can produce errors with idiomatic or rare constructs.
Examples	SYSTRAN, Apertium	Google Translate, (NMT era), DeepL Translator, OpenAI GPT-based translation (A division of Assistive Technology).

Indigenous languages originate from a particular cultural or ethnic community and serve as carriers of cultural identity, heritage, and traditional knowledge. These languages often have unique grammatical structures, phonological systems, and cultural expressions that differ significantly from global languages, for example, English. According to UNESCO (2019), indigenous languages are vital for preserving cultural diversity and traditional knowledge systems. However, many of these languages are underrepresented in digital technologies, including machine translation systems. Thiong'o (1986) emphasizes that indigenous languages are essential for cultural identity and intellectual expression, arguing that language is deeply tied to a community's worldview and cultural practices. Furthermore, linguists such as Crystal (2000) highlights that globalization and technological dominance of major languages have contributed to the marginalization of indigenous languages in digital communication platforms. Understanding indigenous languages is essential in examining how translation technologies handle linguistic diversity.

Linguistic accuracy refers to the degree to which a translation correctly represents the grammatical, semantic, and contextual meaning of the source text in the target language. It involves maintaining correct syntax, vocabulary choice, idiomatic expressions, and overall coherence. According to House (2015), translation accuracy is determined by how well the target text preserves both the semantic meaning and pragmatic function of the original message. Accurate translation requires sensitivity to linguistic structures as well as communicative intent. Similarly, Munday (2016) notes that linguistic accuracy is a key criterion in evaluating translation quality, particularly in comparative studies between human and machine translation outputs. In machine translation, accuracy is often measured through computational metrics such as BLEU scores, which compare machine-generated translations with human reference translations. However, such metrics may not fully capture cultural nuances or contextual appropriateness. Linguistic accuracy is central to assessing the effectiveness of both machine translation tools and human translation when applied to indigenous languages. Artificial Intelligence (AI) refers to the development of computer systems capable of performing tasks that normally require human intelligence, such as language understanding, pattern recognition, and decision-making. In translation technology, AI enables systems to learn linguistic patterns from vast datasets and improve translation quality over time. Russell and Norvig (2010) define artificial intelligence as the study of intelligent agents that perceive their environment and take actions to achieve specific goals. In language processing, AI techniques enable machines to analyze syntax, semantics, and context. Advancements in deep learning have significantly improved translation technology. Neural networks can process entire sentences and capture contextual relationships between words, making translations more natural and fluent. However, AI systems still face limitations when dealing with languages that have limited digital resources or complex cultural expressions. As a result, the development of AI-based translation systems for indigenous languages remains a significant challenge.

Machine Translation Technology (MTT) has evolved through three major phases starting from the 1950s and the 1960s. It began with rule-based systems in the 1950s and 1960s, such as the Georgetown-IBM experiment and SYSTRAN, that relied on linguistic rules. In the 1990s, statistical machine translation (SMT), exemplified by IBM's Candide and later Google Translate, introduced data-driven methods using large corpora. Again, from the 2010s onward, neural machine translation (NMT) systems like DeepL significantly improved translation quality through deep learning. The 2020s have further advanced MTT with large language models (LLMs), such as GPT and Meta's NLLB, enabling more context-aware, multilingual translation. Overall, MTT has progressed from rule-based to statistical, and now to neural and LLM-based approaches.

Empirical Review

The review of relevant literature is organized thematically to highlight methodologies, findings, and the gaps relevant to the present study. Hutchins (2003) conducted a historical analysis of machine translation systems. The researcher compared early rule-based models with more advanced systems. Using a comparative methodology, the researcher evaluated translation outputs across different periods. The findings of the study revealed that although machine translation systems have improved significantly,

they still exhibit inconsistencies in handling context and complex linguistic structures. Hutchins recommended continuous development of MT systems to better address contextual and semantic challenges. This study is relevant to the present research as it establishes that translation accuracy remains a persistent issue, especially for culturally rich languages and multilingual societies.

Kastberg et al. (2012) examined methods for evaluating translation quality using both qualitative and quantitative approaches; their methodology involved combining human assessments with automated evaluation metrics. The researchers found that no single evaluation method is sufficient for assessing translation accuracy. Instead, a hybrid approach yields more reliable results, especially when it involves indigenous languages. The authors recommended integrating human judgment into machine translation evaluation processes for accuracy and proper understanding. This is particularly relevant to the present study, which involves assessing AI translations of Nigerian indigenous expressions that require cultural interpretation of sentences from indigenous languages namely: Igbo, Hausa and Yoruba Languages.

Briggs (2018) investigated the effectiveness of AI-based translation systems, particularly neural machine translation, using an experimental approach. The study compared machine-generated translations with human translations. The findings indicated that AI performs well in translating general texts but struggles with idiomatic expressions, proverbs and culturally specific content. The researcher recommended improving AI systems by incorporating culturally diverse datasets. The recommendation of Briggs (2018) aligns closely with the present study, which focuses on Nigerian indigenous expressions known for their idiomatic, proverbs and cultural complexity.

Nzuanke et al. (2018) explored the challenges of translating Nigerian indigenous languages using computational tools, the study adopted a case-study methodology, analyzing translation outputs between Nigerian languages and English. The findings of the study revealed that machine translation systems perform poorly when applied to Nigerian languages due to limited linguistic resources, structural differences, and cultural nuances. The researchers recommended the development of localized corpora and language-specific AI models.

Fitria (2021) evaluated the accuracy of machine translation tools using error analysis and accuracy scoring. The study compared machine translations with human translations to identify discrepancies. The findings of the study showed that machine translation tools often produce lexical, grammatical, and semantic errors. Fitria recommended the use of human post-editing to improve translation quality. The present study is also interested in identifying inaccuracies in AI-generated translations of indigenous expressions of selected Nigerian indigenous Languages.

Koehn (2010) provided a comprehensive analysis of statistical machine translation, focusing on probability-based models, using computational modeling, the study demonstrated how SMT systems generate translations based on large corpora. The findings of the study indicated that while SMT improves translation efficiency, it struggles with contextual meaning and discourse-level interpretation. Koehn recommended transitioning to more advanced models, such as neural machine translation. The present study provides a theoretical foundation for understanding the capabilities and limitations of AI translation systems.

Chatzikoumi (2019) conducted a systematic review of machine translation evaluation methods, the study analyzed both human and automated evaluation techniques. The findings of the research emphasized the superiority of human evaluation in assessing meaning and fluency. The researcher recommended combining human and automated approaches for more accurate results. This is relevant to the present study, which requires evaluating translations of culturally embedded expressions of selected indigenous Languages in Nigeria.

Han (2022) reviewed modern machine translation evaluation techniques, focusing on neural approaches, using a survey-based methodology, the study examined various evaluation metrics. The findings of this research highlighted the importance of multi-dimensional evaluation frameworks that consider accuracy, fluency, and contextual appropriateness. The researcher recommended adopting comprehensive evaluation models. This is particularly useful to the present study, which involves assessing AI translation accuracy across multiple dimensions of selected indigenous languages.

The studies reviewed reveal some key insights which include the notion that AI translation systems have significantly improved but still struggle with contextual and cultural meanings (Hutchins, 2003; Briggs, 2018). Some of the studies (Kastberg et al., 2012; Chatzikoumi, 2019) posit that translation accuracy requires both human and automated methods. Furthermore, Nigerian indigenous languages are underrepresented in AI models, leading to poor translation accuracy (Nzuanke et al., 2018). In the same vein, machine translation tools (MTT) frequently produce lexical, grammatical, and semantic errors (Fitria, 2021). These are significant contributions, nonetheless, there is a noticeable gap in research focusing specifically on the accuracy of AI in translating selected Nigerian indigenous language expressions into English. Most studies address general machine translation or major global languages, leaving indigenous African languages underexplored, the gaps in the literature underscores the need for focused research on how MTT influence the quality of English translations in various contexts. This study would provide a nuanced understanding of whether these tools enhance or compromise translation standards, addressing an area of inquiry that has yet to receive direct and systematic attention.

Theoretical Framework

This study drew on the Data-Driven Translation Theory (Koehn, 2010) for its theoretical framework. This theory is primarily associated with Philipp Koehn's work, particularly his 2010 book *Statistical Machine Translation*, which focuses on leveraging large bilingual corpora to develop translation models. This approach revolutionized the field of machine translation (MT) by shifting the focus on rule-based methods to statistical models. Again, this theory is primarily associated with statistical machine translation (SMT) and neural machine translation (NMT), which utilize large corpora to improve translation accuracy. Summarily, Philipp Koehn's Data-Driven Translation Theory marked a pivotal shift in translation studies, paving the way for modern neural approaches and contributing to the rapid expansion of multilingual content accessibility. The theory emphasizes that: translation should be based on real linguistic data (corpora) rather than handcrafted grammatical rules, patterns of translation are learned automatically from large collections of bilingual texts and translation quality improves as

the quantity and quality of data increase. The core principles of the theory are: Empiricism over Rules. Traditional translation theories relied on linguistic rules and human expertise. In contrast, Koehn's model argues that: Language patterns can be discovered from data and translation equivalence emerges from frequency and probability. The theory is relevant to the study say by stating that the study draws on this theory, it implies that: the research is grounded in empirical analysis of language data, it uses machine translation tools in translation and the study prioritizes observable translation patterns over abstract theorizing. The strengths of the Framework include objectivity, scalability, real-world relevance and technological alignment. By objectivity, this research reduces subjective human bias in translation analysis. By scalability this research can handle large volumes of text efficiently. By real-world relevance, this research reflects how language is actually used and by technological alignment, it supports AI-driven translation systems. Despite its strengths, the theory has some drawbacks: poor results when data is limited or biased, cultural and contextual meanings may be overlooked, above all, it focuses more on patterns than meaning. In conclusion, using Data-Driven Translation Theory (Koehn, 2010) as a theoretical framework means the study adopts a modern, computational, and empirical approach to translation, where meaning and equivalence are derived from large-scale language data rather than fixed linguistic rules. It aligns the study with contemporary developments in artificial intelligence and digital linguistics, making it particularly suitable for research in machine translation and digital communication.

Methodology

This study focuses on the translation of some Igbo, Hausa and Yoruba Language sentences to English language comparing Machine Translation Tools (MTT) and Human Translator (HT). Some Igbo, Hausa and Yoruba language sentences incorporating proverbs, idioms and cultural values were drawn from the fields of education, healthcare, legal, business and international diplomacy. These areas are viewed as the critical area of human endeavours.

The data used in this research are the sentences collected randomly. The MTT such as Google Translate, DeepL Translator, OpenAI GPT were downloaded and used for the translations. The human translators who are experts in the field of translation are also deployed for the study. The specimen sentences which are the original English sentences from where the researchers generate the source sentences are given to the MTT and HTs to translate.

The population of 200 level students were carefully chosen because they are conversant with their environment. Reasons Why the Researchers Used These Tools (Google Translate, DeepL Translator, OpenAI GPT). Comparison of Translation Quality: The research aimed to analyze how different MTT handle translation accuracy, context, and idiomatic expressions. Coverage of Multiple Languages: The selected tools support a wide range of languages, allowing for diverse linguistic comparisons. Evaluation of AI Translation Capabilities: The study sought to examine the impact of NMT models on translation quality. Assessment of User Accessibility: These tools are widely used by individuals and organizations, making them relevant for practical research. Understanding Contextual Accuracy: The study focused on how well these tools maintain

meaning, cultural nuances, and grammatical correctness in translations. This comparative analysis helps in understanding the strengths and limitations of AI-powered translation tools in different linguistic settings.

Data Analysis

The study examined translation of the indigenous language using three major MTT, each of these tools employs different AI techniques, including neural networks and deep learning, to improve translation accuracy and efficiency.

1. Challenge in Expression of Cultural Nuances

The MTT translation of the indigenous language expressions to English showed challenges in relation to appropriate expression of cultural nuances, proverbs, idiomatic expressions, and context-specific meanings. The fragments analysed below are examples illustrating these challenges from the samples translated.

Extract 1: Igbo Expression

‘Ma obu si ya **ndo**’.

The Igbo expression "ndo" could be literally translated as "sorry". However, in the context of Igbo culture, "*Ndo*" is used not just for apologizing but also as an expression of sympathy, consolation, or concern. For example, if someone trips and falls, an Igbo speaker might say "*Ndo*," meaning "*I sympathize with you*," not "*I apologize*." In the translation above, machine translation: ‘A wise teacher does not need sorry’ in this case misses these cultural nuances. For the machine translator, the expression simply means that a wise teacher knows their job and carries it out without fear or pity, not necessarily needing to be told 'sorry'.

Extract 2: Hausa Expression

‘Babu mai *fushi* da ke’.

Translation is often more than substituting one word for another; it requires an understanding of culture, tone, and use. The machine translation of the Hausa word ‘fushi’, as ‘angry’ demonstrates the need for an understanding of the culture, tone and contextual use of language. The translated word ‘angry’ is a direct English rendering of ‘fushi’ and could be inadequate in some contexts. Within Hausa culture, *fushi* is not confined to the expression of anger or irritation. While it could be used to describe displeasure, it also appears in lighter, more relational contexts. In particular, it can function much like ‘wasa’, the Hausa word for *play* or *jest*. In these situations, *fushi* does not convey hostility, but instead reflects mock seriousness, teasing, or playful interaction among people who are familiar with one another. This dual usage places *fushi* on a spectrum. At one end, it may indicate genuine annoyance; at the other, it is a tool for banter that strengthens social bonds rather than damages them. To reduce it only to anger is to strip away this cultural flexibility of the use of the word. The meaning of *fushi* is therefore context-dependent. It is shaped by relationship, tone, and intent, and at times

it aligns more closely with *play* than with *anger*. A culturally accurate translation must therefore account for both its emotional and its social dimensions.

Extract 3: Yoruba Expression

‘Ojọ́ a bí ni kọ́ ní à ń dàgbà’.

Contextually in Yorùbá culture, “**ojọ́ a bí ni kọ́ ní à ń dàgbà**” is used not just an ordinary expression but it is loaded with deep meanings. Yoruba speakers may use this sentence for encouragement. On the other hand, it may be used as a proverbial means to teach patience and endurance. However, the MTT translation for the expression: “learning is a process of growth that takes time.” is a straight forward interpretation that misses out on other semantic possibilities that that could be tied to the expression depending on the context of use.

2. Challenge in Proverb and Idiom Translation

Proverbs and idioms are also areas in language use where machine translation demonstrated some inadequacies in relation to expression of contextual meaning. Extracts 4, 5 and 6 below demonstrate machine’s limitation at reflecting cultural nuances in English translations of indigenous languages’ proverbs and idioms.

Extract 4: Igbo Proverb

‘énwèrè íké ì kpuga ìnyinya nà íyí mànà enweghì íké ì me yá kà ọ ñụọ mmiri n’íké’ [Literal machine translation: A man can take a horse to the stream but cannot force it to drink water].

The meaning of this proverb in Igbo is that a man can provide everything for his children but cannot force them to be good, rather it is a matter of choice. In the case of a teacher, the teacher cannot force the students to obey and learn, rather it depends on the students’ willingness to learn. In the MTT version, the implied wisdom behind the words is not reflected; engendered by contextual influence is lost.

Extract 5: Hausa Proverb

“*Lafiya uwar jiki, babu mai fushi da ke.*” [Literal machine translation: Good morning mother, no one is angry with you]

The above Hausa proverb means that *health is wealth*, and it is something no one takes lightly. However, in the MTT version, translating “mother” as *uwa* does not reflect the true cultural expression in Hausa.

Extract 6: Yoruba Proverb:

‘**Aşo tí a bá fí pẹ̀lẹ́ tú ní àsìkò, kì í fàya.**’

(Literal machine translation: ‘A cloth that is mended early will not tear apart).

In Yoruba language, the meaning of this proverb could be extended to different cases. For example, in health matters, the proverb could suggest that proactive approaches to minor ailments prevent them from escalating to a health threatening situation. In the MTT version, this semantic extension is lost, and the implied wisdom behind the proverb is also not reflected.

3. Challenge with Context Specific Lexical Meaning in Translation

The meanings that could be assigned to words are determined by the linguistic environment, cultural differences, possibility of a word being polysemous. Extracts 6, 7 and 8 analysed below reflect machine's inability to reflect these shades on meanings in its translation of indigenous language lexis to their English equivalent.

Extract 7: Igbo word: 'Akwukwọ.

The word "*akwukwọ*" has the following possible machine translations: "Book" (Correct if it refers to school materials), "Leaf" (Correct in botanical contexts), "Paper" (Correct in printing or administrative contexts), "Certificate" (Correct in academic or official contexts). However, it is important to note that if the MTT does not recognize the context of use, it may translate "*akwukwọ*" incorrectly, leading to confusion. For instance, "*akwukwọ ndụmọdụ*" means advisory letter, not advice book as translated by the MTT.

Extract 8: Hausa word: 'uwa'

The Hausa word *uwa* has the following possible machine translations: biological mother, adoptive mother, stepmother, and unconditional support contexts. Nevertheless, If the MTT ignores context, it may mistranslate *uwa* and cause confusion. *Uwa* has multiple meanings (biological mother, adoptive mother, stepmother, unconditional support), and MTT can translate it inaccurately if translation is not done with the context of use in view.

Extract 9: Yoruba word: 'òdodo'

This word is one of the Yoruba words that are semantically ambiguous. For example, *ododo* can be translated in the following ways: Natural truth, transparency, sound judgement, righteousness, and authenticity. However, machine translation rendered it literally as justice and flower. It must be pointed out that this challenge in content-specific meaning of Yoruba words may still linger in Yoruba/English machine translation because many of Yoruba lexical items are polysemous in meaning.

Discussion of Findings

This section discusses the findings of the study based on the analysis of samples examined. The original Igbo source sentence, 'Ónyé mára íhé màrà nà énwèrè íké ì kpuga ìnyịnya nà íyí màrà enweghị íké ì me yá kà ọ ñụọ mmiri...' (**Extract 4**) contains a proverb-like structure equivalent to the English idiom 'you can lead a horse to water, but you cannot make it drink'. This figurative expression serves as the foundation for evaluating machine translation accuracy. OpenAI ChatGPT demonstrates a moderate

level of accuracy. For example, it renders the idea as ‘He who knows the way can ride a horse to the stream but may not have the strength to make it drink’. While the meaning is partially preserved, the metaphor ‘ride a horse to the stream’ deviates from the intended meaning of ‘lead a horse,’ indicating lexical inaccuracy in the machine English translation. However, ChatGPT performs better in interpreting the pedagogical extension of the proverb, as seen in ‘good teachers know how to make knowledge accessible to their students,’ which reflects an attempt to contextualize the metaphor within educational setting. Despite this, the concluding clause ‘A wise teacher does not need sorry in this case as good mothers’ is grammatically flawed and semantically unclear.

Google Translate (40%) performs poorly in both semantic and syntactic accuracy. For instance, the translation ‘there is a way to lead a horse and there is no way to make it drink water in the forest’ introduces the word “forest,” which is absent in the source text, thereby distorting the original meaning. Additionally, the phrase ‘the way to drink water is to drink a horse’ is a clear mistranslation, demonstrating the system’s inability to process figurative language. The closing statement ‘No sorry for the teach in the case with mother’ further highlights issues of grammatical incoherence and loss of meaning. DeepL Translator could not be evaluated because the Igbo language is not supported on the platform. This limitation underscores the broader issue of underrepresentation of African languages in advanced machine translation systems. In contrast, the Human Translation provides a more accurate and culturally appropriate rendering. For example, ‘you can take a horse to the stream but you can’t force it to drink’ correctly captures the idiomatic meaning of the source text. Furthermore, the translator effectively conveys the educational metaphor in ‘some skillful/good teachers know how to make that horse to thirst for water,’ preserving both meaning and stylistic nuance. The concluding statement, ‘a wise teacher knows his job just like a good mother,’ reflects a culturally grounded and semantically coherent interpretation of the original message.

The Hausa source sentence, ‘Sakaci wajen ganin likita yakan jawo da na sani, domin bashin kula da kananan cututtuka na iya jawo rashin lafiya. Lafiya uwar jiki, babu mai fushi da ke,’ (**Extract 5**) conveys a culturally grounded warning about neglecting health, emphasizing that carelessness in treating minor illnesses can result in serious consequences. It also employs the proverb “*Lafiya uwar jiki*” (health is the mother/keeper of the body), which requires interpretive sensitivity with precision. OpenAI ChatGPT shows a relatively high level of semantic accuracy. For instance, it translates the first clause as ‘Neglecting to see a doctor often leads to regret, because ignoring minor ailments can result in serious illness,’ which closely aligns with the original meaning of ‘Sakaci wajen ganin likita yakan jawo da na sani...’. This reflects a good understanding of the causal relationship expressed in the source text. Furthermore, the rendering of ‘Health is the body’s greatest wealth’ shows an attempt to interpret ‘Lafiya uwar jiki’ metaphorically rather than literally. However, while this interpretation is contextually acceptable, it slightly shifts the meaning from “mother/guardian of the body” to “wealth,” thereby losing some cultural specificity. The concluding clause ‘no one gets angry with good health’ is a reasonable approximation of ‘babu mai fushi da ke,’ although it appears somewhat literal and less idiomatic in the English translated version.

Google Translate, on the other hand, exhibits significant semantic and pragmatic weaknesses. For example, the translation ‘Delaying seeing a doctor can lead to illness, I

know...' introduces the phrase 'I know' which is absent in the original Hausa sentence, thereby distorting the intended meaning. Additionally, the repetition 'because ignoring minor ailments can lead to illness' reflects redundancy and lack of stylistic refinement. More critically, the phrase 'Good morning, mother, no one is angry with you' is a clear mistranslation of 'Lafiya uwar jiki, babu mai fushi da ke'. The insertion of "Good morning" demonstrates a failure to correctly interpret the lexical item 'uwar' (mother), leading to a completely irrelevant output. This indicates a breakdown in both lexical disambiguation and contextual understanding. DeepL Translator produces an identical output to Google Translate in this case, suggesting either reliance on similar translation patterns or limitations in processing Hausa input effectively. Like Google Translate, it fails to capture the meaning of the proverb and introduces irrelevant expressions such as 'Good morning, mother'. This further underscores the inadequacy of current MTT systems in handling Hausa language data and culturally embedded expressions. In contrast, the Human Translation (HT) provides a more accurate and culturally sensitive rendering of the source text. For instance, 'Carelessness in consulting a doctor can lead to regrets; because trivializing symptoms can lead to serious illness' effectively captures the cause-and-effect relationship in the original sentence. The translation of 'Lafiya uwar jiki' as 'Health is the keeper/mother of the body' preserves the metaphorical and cultural significance of the expression. Additionally, the clarification 'no one hates you (health)' demonstrates an interpretive effort to make implicit meaning explicit for the target audience while maintaining fidelity to the source text.

The Yoruba source sentence, 'Ètò ẹ̀kọ́ dà bí igi gbígbin: bí gbòngbò rẹ̀ bá ẹ̀ jinlẹ̀ sí, ni ẹ̀ka rẹ̀ yóò ẹ̀ lagbara sí...', is fundamentally educational rather than legal in orientation. It uses an extended metaphor comparing education to a growing tree, emphasizing patience, gradual development, and strong foundational learning. This creates an immediate mismatch with the English specimen sentence, which is rooted in legal discourse (justice, truth, and honesty). This misalignment significantly affects translation evaluation. OpenAI ChatGPT attempts to align the output with the legal tone of the specimen sentence rather than the educational content of the source. For example, it produces 'Slowly but surely, the wheel of justice turns, and it will eventually crush everything that is unjust'. This reflects the original English specimen rather than the Yoruba source text. Similarly, 'If you build your argument on lies, it will collapse like a house built on sand' mirrors the legal metaphor of falsehood and justice but does not correspond to the Yoruba idea of *gradual learning and patience*. While the translation is fluent and coherent in English, it demonstrates a failure of fidelity to the source text, indicating that the system prioritizes contextual guessing or alignment with expected themes over accurate translation.

Google Translate exhibits a similar pattern of deviation. Its rendering, 'The wheels of justice turn slowly, but all that is unjust will be overthrown,' again reflects the legal specimen rather than the Yoruba source. Like ChatGPT, it introduces 'house built on sand' and 'honesty is the surest principle,' none of which appear in the Yoruba text. This suggests that the system is not processing the input text faithfully but instead reproducing familiar English idioms associated with justice. Although grammatically sound, the translation is semantically inaccurate in relation to the source. DeepL Translator performs the weakest among the MTTs in this case. Its output, 'This is a matter of great

importance; we must ensure that everything is done properly to avoid any misunderstandings...’, is entirely unrelated to both the Yoruba source and the English specimen. Furthermore, the continuation ‘If you do not know the way, you will end up in the wrong place’ and ‘the path is clear, for the light has shone upon it’ introduces new ideas that are absent from the source text. This indicates a complete breakdown in translation accuracy, likely due to limited or non-existent support for Yoruba language input. The human translation, although more aligned with the legal specimen sentence than the Yoruba source, demonstrates better interpretive coherence than the MTTs. For example, ‘The bicycle (wheels) of Justice/judgement turns slowly, but breaks all things that are not true or just’ attempts to preserve the metaphor of justice while maintaining clarity. Similarly, ‘if you gather your riches or asset on lies it will crush... like a house built on sand’ reflects the consequences of dishonesty, consistent with the specimen sentence. However, like the MTTs, the HT does not accurately represent the educational metaphor in the Yoruba source text. Instead, it adapts the translation to fit the legal theme, indicating a shift from direct translation to contextual reinterpretation.

Conclusion

MTT have revolutionized language translation, offering fast and cost-effective solutions. These tools, including OpenAI Chat GPT, Google Translate, and DeepL Translator, rely on AI and NMT models to process and generate translations. While MTT enhance accessibility and efficiency, their impact on translation quality remains a topic of debate. Key concerns include accuracy, contextual appropriateness, fluency, and cultural sensitivity. Research indicates that while MTT perform well with common language pairs and simple texts, they struggle with idiomatic expressions, domain-specific terminology, and nuanced linguistic structures. Human translators still play a crucial role in post-editing MTT outputs to ensure quality, accuracy, and coherence. MTT should continue to evolve by incorporating more advanced neural network models and deep learning techniques to improve contextual understanding. For example, Google Translate has improved significantly with NMT, but it still struggles with idiomatic expressions. Developers should focus on refining algorithms to better detect and process idiomatic phrases, cultural nuances, and syntactic complexities.

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Appendix A: Specimen Sentences, Source Sentences, MTT and HT Translations

Table 2 Igbo Language

Specimen sentences	Source sentences	MTT and HT
Education: Specimen sentence	Source sentence Igbo Language	MTT and HT
1. "A wise teacher knows that you can lead a horse to water, but you can't make it drink. However, the best teachers make the water so inviting that the horse thirsts for it. This is not the case of a sorry situation; a wise teacher knows the job like a good mother"	"Ọnyé mára íhé màrà nà énwèrè íké ì kpuga ìnyinya nà íyí màrà enweghị íké ì me yá kà ọ ñụọ mmiri n'íké. Étú osinà dī, ézígbò ndi nkúzí màrà ètù ési èmè kà yá bú mmiri guọ yá bú ìnyinya águu. Nke abuhi imere onye nkuzi mara ihe ebere, ma obu si ya ndo dika ezigbo nne"	OpenAI Chat GPT ^{60/100%} "He who knows the way can ride a horse to the stream but may not have the strength to make it drink. Likewise, good teachers know how to make knowledge accessible to their students. A wise teach does not need sorry in this case as good mothers"
		Google Translate ^{40/100%} A person who knows the world knows that there is a way to lead a horse and there is no way to make it drink water in the forest. However, the wise man knows that the way to drink water is to drink a horse. No sorry for the teach in the case with mother"
		DeepL Translator ^{x/100%} The Igbo language is not included in the DeepL Translator Application software.
		HT ^{80/100%} "A wise man knows that you can take a horse to the stream but you can't force it to drink. However, some skillful/good teachers know how to make that horse to thirst for water. In this situation, a wise teacher knows his job just like a good mother"

Table 3: Hausa Language

Specimen sentences	Source sentences Hausa	Machine (AI) and Human Translations
Health: Specimen sentence	Source sentence Hausa	Machine (AI) and Human Translations
1. "An apple a day keeps the doctor away, but ignoring small ailments today may bring a storm tomorrow. Take care of your body, for it is the only house you cannot move out of."	"Sakaci wajen ganin likita yakan jawo da na sani, domin bashin kula da kananan cututtuka na iya jawo rashin lafiya. Lafiya uwar jiki, babu mai fushi da ke."	Open Chat GPT "Neglecting to see a doctor often leads to regret, because ignoring minor ailments can result in serious illness. Health is the body's greatest wealth, and no one gets angry with good health."
		Google Translate "Delaying seeing a doctor can lead to illness, I know, because ignoring minor ailments can lead to illness. Good morning, mother, no one is angry with you."

		DeepL Translator "Delaying seeing a doctor can lead to illness, I know, because ignoring minor ailments can lead to illness. Good morning, mother, no one is angry with you."
		Human Translator "Carelessness in consulting a doctor can lead to regrets; because trivializing symptoms can lead to serious illness. Health is the keeper/mother of the body, no one hates you (health)".

Table 4: Yoruba Language

Law: Specimen sentence	Source sentence Yoruba Language	MTT and HT
1. "The wheels of justice turn slowly, but they grind exceedingly fine. If you build your case on lies, it will crumble like a house of cards. Speak the truth and stand on solid ground, for honesty is the best policy."	Ètò ẹ̀ kọ́ dà bí ọ̀gi gbígbin: bí gbòngbò ẹ̀ bá ẹ̀ jinlẹ̀ sí, ní ẹ̀ ká ẹ̀ yóò ẹ̀ lagbara sí. Bí a bá kánju fúnrúgbìn ilànà ẹ̀kọ́, ẹ̀so tí a ọ̀ kórè lẹ̀ parun kíákíá. A gbòdọ̀ fi sùúrù kọ̀ nì, kí a sì kọ̀ nì lẹ̀ kọ̀ ní pẹ̀ lẹ̀ pẹ̀ láìsẹ̀mẹ̀ lẹ̀, nítorí ọ̀jọ̀ a bí nì kọ̀ nì à ñ dàgbà."	OpenAI Chat GPT "Slowly but surely, the wheel of justice turns, and it will eventually crush everything that is unjust. If you build your argument on lies, it will collapse like a house built on sand. Speak the truth and stand firmly on it like a solid foundation, for honesty is the most reliable principle of all."
		Google Translate "The wheels of justice turn slowly, but all that is unjust will be overthrown. If you build your case on lies, it will crumble like a house built on sand. Speak the truth, stand on it as a good foundation, for honesty is the surest principle."
		DeepL Translator "This is a matter of great importance; we must ensure that everything is done properly to avoid any misunderstandings. If you do not know the way, you will end up in the wrong place. 'In truth, the path is clear, for the light has shone upon it, and the truth is that the way is now open.'"
		HT The bicycle (wheels) of Justice/judgement turns slowly, but breaks all things that are not true or just, if you gather your riches or asset on lies it will crush it is like a house built on sand. Say the truth, stand on it like a solid foundation, because saying the truth is the best legacy.